

ArrestShield ML Status

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This file is the single source of truth for project status.

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The earlier hand-written PDF (12 August 2026) claimed "ML COMPONENT - Complete" on its front page while its own page 4 listed the reasons production was blocked. Generating the PDF from this file removes that failure mode by construction.

Last verified: 13 August 2026.

One-sentence status

The software pipeline is complete and works end to end; the detector is not accurate enough to deploy, and the reason is the training data, not the model.

What is complete

Component	State	Evidence
Dataset pipeline	Complete	52,206 conversations, 615,084 turns, zero split-group leakage
Baseline + ladder models	Complete	SGD and XGBoost, seeds 17/42/93
Multi-task auxiliary heads	Complete	Scam type, stage, 9 tactics
Risk fusion	Complete	XGBoost over out-of-fold base scores
ASR	Complete	Local Whisper-tiny; 0.00 WER on one English clip
Inference API	Complete	/healthz, /v1/model, /v1/detect/text, /v1/detect/audio
LLM honeypot	Complete	Signed handoff, default-deny gate, live Groq engagement verified
Audio -> honeypot chain	Complete	scripts/run_honeypot.py --audio
Tests	90 passing	python -m pytest tests/ -q

Verified end to end on 13 August 2026:

```
`audio -> Whisper -> transcript -> detector -> HMAC-signed handoff -> eligibility gate
-> live Groq persona reply -> non-evidential transcript`
```

What is not complete

Limitation	Measurement	Consequence
False positives on financial dialogue	18.51% FPR on 13,071 unseen banking conversations	Unusable in the deployment domain
Threshold instability	Seeds chose 0.657 / 0.896 / 0.879 -> 5,698 / 813 / 810 false positives	Operating point is not reproducible
Score calibration	3s sine tone scores 0.9473 vs 0.9479 threshold	Score is a style detector, not a probability
Hinglish detection quality	Held-out-source ROC-AUC 0.756 +/- 0.055	Better than 0.550, still not deployable
Hinglish evaluation size	21 held-out negatives	Hinglish FPR is not a measurable quantity
Positive label quality	All 2,256 positives source-silver; 71% synthetic	No real-call accuracy claim is possible
Hinglish ASR	Never tested	Only an English clip has been transcribed
ASR speed	38s for 3s of audio on CPU	File upload only; not live monitoring
Human gold set	0 of 150 collected	Final promotion gate unavailable

Corrected headline numbers

Do not quote the 98.74% F1 from reports/baseline_v1/. It is a random-split number produced under a corpus shortcut and is not a performance claim.

The defensible numbers are:

- **Held-out-source Hinglish ROC-AUC: 0.756 +/- 0.055** (3 seeds, threshold-free)
- **Unseen English FPR: 18.51%** on banking dialogue, 5.06% aggregate over 48,216
- **Synthetic English recall: 1.000** - not evidence of anything

The corpus finding

indian_cyber_scam_phonecall_hinglish ships 10,000 rows. **9,257 are exact duplicates.** 743 unique conversations survive: 633 scam, 110 legitimate.

The Source counts table in docs/datasets/CANONICAL_BUILD_REPORT.md lists input counts, before the 10,140 exact duplicates were removed. Any figure derived from that table overstates the Hinglish corpus by roughly 13x.

The shortcut, and the fix that worked

49,950 of 52,206 conversations come from three corpora containing only legitimate examples. "Does this look like banking77 / DailyDialog / schema-guided?" was therefore close to a perfect label, and the detector learned it.

Restricting training to sources containing **both** labels removed the shortcut by construction:

Held-out source	Full corpus (36,101 train)	Mixed-only (2,368 train)
indian_cyber_scam_hinglish	0.550 +/- 0.043	0.756 +/- 0.055
synthetic_multi_agent	0.979 +/- 0.009	1.000 +/- 0.000
synthetic_scam_dialogue	0.971 +/- 0.012	1.000 +/- 0.000

Deleting 94% of the training data improved held-out Hinglish AUC by 0.21, roughly four times the seed noise. Reproduce with

```
python scripts/train_mixed_source_detector.py.
```

Approaches tested and rejected

Both were expected to help and did not. They are recorded because a negative result from a controlled experiment is evidence.

Approach	Result	Why it was dropped
LLM-generated Hinglish augmentation (Groq)	AUC 0.756 -> 0.743	No gain at n=8; a third synthetic style does not add information
External BFSI Hinglish hard negatives (2,452 rows)	AUC 0.756 -> 0.615	Single-label source reintroduced the shortcut it was meant to remove

The second is the more instructive: adding legitimate in-domain conversation *hurt*, because it arrived as a source containing no scam examples. The requirement is legitimate financial conversation from a source that also contains scams - which is what collection has to provide.

Boundary guarantee

No LLM produces a feature, score, threshold, or label anywhere in detection.

- `src/arrestshield/honeypot.py` imports no detector module; a test enforces it.
- The honeypot requires an HMAC-signed detector event and cannot be reached otherwise.
- Honeypot transcripts are stamped `excluded_from_detector_training: true`.
- Whisper is a trained ASR model, not an LLM, and produces words only.
- LLM-generated training rows are provenance: `llm_synthetic`, train-split only, weighted 0.5, and structurally barred from validation and test.

How to verify on this PC

```
.venv\Scripts\python.exe -m pytest tests\ -q
.venv\Scripts\python.exe scripts\train_mixed_source_detector.py
.venv\Scripts\python.exe scripts\evaluate_unseen_english.py
.venv\Scripts\python.exe scripts\run_api.py
.venv\Scripts\python.exe scripts\run_honeypot.py --demo --research-mode
```

The honeypot requires `GROQ_API_KEY` in a git-ignored `.env`. Without it every other command still runs.